

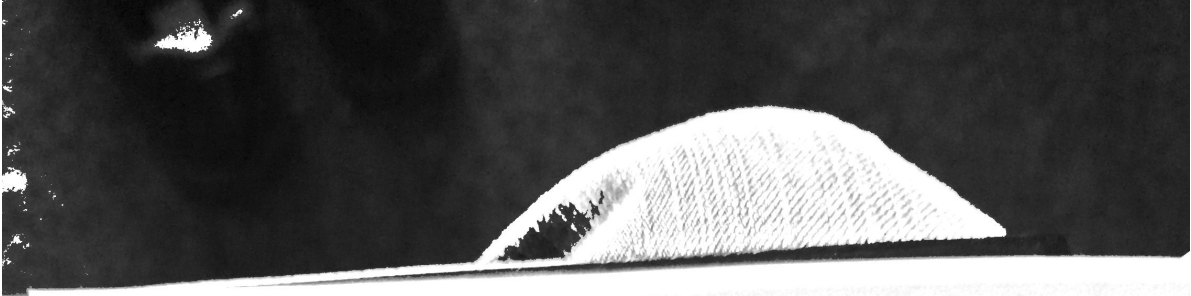
Product Specifications



Courtesy of Specialized Bicycle Components

EXHIBIT 6-1

One of Specialized's existing mountain bikes with a suspension fork.



Specialized Bicycle Components was interested in developing a new front suspension fork for the mountain bike market. Although the firm was already selling suspension forks on its bicycles (Exhibit 6-1), it was interested in exploring designs that would provide higher value for the recreational cyclist.

The development team had spent a great deal of time identifying customer needs. In addition to logging many hours of riding on suspended bikes themselves, the members of the team had interviewed lead users at mountain bike races and recreational cyclists on local trails, and they also had spent time working with dealers in their stores. As a result of this process they had assembled a list of customer needs. They now faced several challenges:

- How could the relatively subjective customer needs be translated into precise targets for the remaining development effort?
- How could the team and its senior management agree on what would constitute success or failure of the resulting product design?
- How could the team develop confidence that its intended product would garner a substantial share of the suspension fork market?
- How could the team resolve the inevitable trade-offs among product characteristics like cost and weight?

This chapter presents a method for establishing product specifications. We assume that the customer needs are already documented as described in Chapter 5, Identifying Customer Needs. The method employs several simple information systems, all of which can be constructed using conventional spreadsheet software.

What Are Specifications?

Customer needs are generally expressed in the “language of the customer.” The primary customer needs for the suspension fork are listed in Exhibit 6-2. Customer needs such as “the suspension is easy to install” or “the suspension enables high-speed descents on bumpy trails” are typical in terms of the subjective quality of the expressions; however, while such expressions are helpful in developing a clear sense of the issues of interest to customers, they provide little specific guidance about how to design and engineer the product. They simply leave too much margin for subjective interpretation. For this reason, development teams usually establish a set of specifications, which spell out in precise, measurable detail *what* the product has to do. Product specifications do not tell the team *how* to address the customer needs, but they do represent an unambiguous agreement on what the team will attempt to achieve to satisfy the customer needs. For example, in contrast to the customer need that “the suspension is easy to install,” the corresponding specification might be that “the average time to assemble the fork to the frame is less than 75 seconds.”

We intend the term *product specifications* to mean the precise description of what the product has to do. Some firms use the terms “product requirements” or “engineering characteristics” in this way. Other firms use “specifications” or “technical specifications” to refer to key design variables of the product such as the oil viscosity or spring constant of the suspension system. These are just differences in terminology. For clarity, let us be

No.	Need		Imp.
1	The suspension	reduces vibration to the hands.	3
2	The suspension	allows easy traversal of slow, difficult terrain.	2
3	The suspension	enables high-speed descents on bumpy trails.	5
4	The suspension	allows sensitivity adjustment.	3
5	The suspension	preserves the steering characteristics of the bike.	4
6	The suspension	remains rigid during hard cornering.	4
7	The suspension	is lightweight.	4
8	The suspension	provides stiff mounting points for the brakes.	2
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5
10	The suspension	is easy to install.	1
11	The suspension	works with fenders.	1
12	The suspension	instills pride.	5
13	The suspension	is affordable for an amateur enthusiast.	5
14	The suspension	is not contaminated by water.	5
15	The suspension	is not contaminated by grunge.	5
16	The suspension	can be easily accessed for maintenance.	3
17	The suspension	allows easy replacement of worn parts.	1
18	The suspension	can be maintained with readily available tools.	3
19	The suspension	lasts a long time.	5
20	The suspension	is safe in a crash.	5

precise about a few definitions. A *specification* (singular) consists of a *metric* and a *value*. For example, “average time to assemble” is a metric, while “less than 75 seconds” is the value of this metric. Note that the value may take on several forms, including a particular number, a range, or an inequality. Values are always labeled with the appropriate units (e.g., seconds, kilograms, joules). Together, the metric and value form a specification. The *product specifications* (plural) are simply the set of the individual specifications.

Specifications Established?

In an ideal world, the team would establish the product specifications once early in the development process and then proceed to design and engineer the product to exactly meet those specifications. For some products, such as soap or soup, this approach works quite well; the technologists on the team can reliably concoct a formulation that satisfies almost any reasonable specifications; however, for technology-intensive products this is rarely possible. For such products, specifications are established at least twice. Immediately after identifying the customer needs, the team sets *target specifications*. These specifications represent the hopes and aspirations of the team, but they are established before the team knows what constraints the product technology will place on what can be achieved.

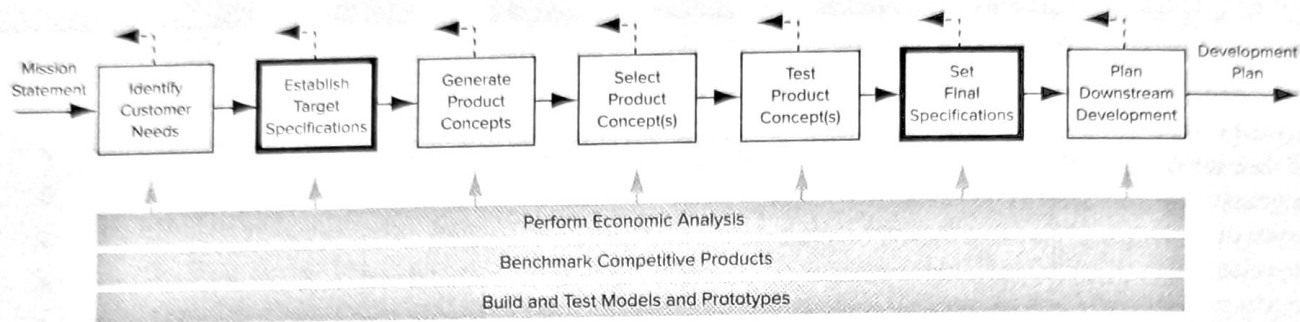


EXHIBIT 6-3 The concept development process. The target specifications are set early in the process, but setting the final specifications must wait until after the product concept has been selected.

The team's efforts may fail to meet some of these specifications and may exceed others, depending on the product concept the team eventually selects. For this reason, the target specifications must be refined after a product concept has been selected. The team revisits the specifications while assessing the actual technological constraints and the expected production costs. To set the *final specifications*, the team must frequently make hard trade-offs among different desirable characteristics of the product. For simplicity, we present a two-stage process for establishing specifications, but we note that in some organizations specifications are revisited many times throughout the development process.

The two stages in which specifications are established are shown as part of the concept development process in Exhibit 6-3. Note that the final specifications are one of the key elements of the development plan, which is usually documented in the project's *contract book*. The contract book (described in Chapter 19, Project Management) specifies what the team agrees to achieve, the project schedule, the required resources, and the economic implications for the business. The list of product specifications is also one of the key information systems used by the team throughout the development process.

This chapter presents two methods: The first is for establishing the target specifications and the second is for setting the final specifications after the product concept has been selected.

Establishing Target Specifications

As Exhibit 6-3 illustrates, the target specifications are established after the customer needs have been identified but before product concepts have been generated and the most promising one(s) selected. An arbitrary setting of the specifications may not be technically feasible. For example, in designing a suspension fork, the team cannot assume in advance that it will be able to achieve simultaneously a mass of 1 kilogram, a manufacturing cost of \$30, and the best descent time on the test track, as these are three quite aggressive specifications. Actually meeting the specifications established at this point is contingent upon the details of the product concept the team eventually selects. For this reason, such preliminary specifications are labeled "target specifications." They are the goals of the development team, describing a product that the team believes would succeed in the marketplace. Later these specifications will be refined based on the limitations of the product concept actually selected.

The process of establishing the target specifications contains four steps:

1. Prepare the list of metrics.
2. Collect competitive benchmarking information.
3. Set ideal and marginally acceptable target values.
4. Reflect on the results and the process.

Step 1: Prepare the List of Metrics

The most useful metrics are those that reflect as directly as possible the degree to which the product satisfies the customer needs. The relationship between needs and metrics is central to the entire concept of specifications. The working assumption is that a translation from customer needs to a set of precise, measurable specifications is possible and that meeting specifications will therefore lead to satisfaction of the associated customer needs.

A list of metrics is shown in Exhibit 6-4. A good way to generate the list of metrics is to contemplate each need in turn and to consider what precise, measurable characteristic of the product will reflect the degree to which the product satisfies that need. In the ideal case, there is one and only one metric for each need. In practice, this is frequently not possible.

For example, consider the need that the suspension be "easy to install." The team may conclude that this need is largely captured by measuring the time required for assembly of the fork to the frame; however, note the possible subtleties in this translation. Is assembly time really identical to ease of installation? The installation could be extremely fast but require an awkward and painful set of finger actions, which ultimately may lead to worker injury or dealer frustration. Because of the imprecise nature of the translation process, those establishing the specifications should have been directly involved in identifying the customer needs. In this way the team can rely on its understanding of the meaning of each need statement derived from firsthand interactions with customers.

The need for the fork to reduce vibration to the user's hands may be even more difficult to translate into a single metric, because there are many different conditions under which vibration can be transmitted, including small bumps on level roads and big bumps on rough trails. The team may conclude that several metrics are required to capture this need, including, for example, the metrics "attenuation from dropout to handlebar at 10 Hz" and "maximum value from the Monster." (The "Monster" is a suspension test developed by *Mountain Bike* magazine.)

A simple needs-metrics matrix represents the relationship between needs and metrics. An example needs-metrics matrix is shown in Exhibit 6-5. The rows of the matrix correspond to the customer needs, and the columns of the matrix correspond to the metrics. A mark in a cell of the matrix means that the need and the metric associated with the cell are related; performance relative to the metric will influence the degree to which the product satisfies the customer need. This matrix is a key element of the *House of Quality*, a graphical technique used in *Quality Function Deployment*, or *QFD* (Hauser and Clausing, 1988). In many cases, we find the information in the needs-metrics matrix is just as easily communicated by listing the numbers of the needs related to each metric alongside the list of metrics (the second column in Exhibit 6-4). There are some cases, however, in which the mapping from needs to metrics is complex, and the matrix can be quite useful for representing this mapping.

Metric No.	Need Nos.	Metric	Imp.	Units
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB
2	2, 6	Spring preload	3	N
3	1, 3	Maximum value from the Monster	5	g
4	1, 3	Minimum descent time on test track	5	s
5	4	Damping coefficient adjustment range	3	N-s/m
6	5	Maximum travel (26-in. wheel)	3	mm
7	5	Rake offset	3	mm
8	6	Lateral stiffness at the tip	3	kN/m
9	7	Total mass	4	kg
10	8	Lateral stiffness at brake pivots	2	kN/m
11	9	Headset sizes	5	in.
12	9	Steertube length	5	mm
13	9	Wheel sizes	5	List
14	9	Maximum tire width	5	in.
15	10	Time to assemble to frame	1	s
16	11	Fender compatibility	1	List
17	12	Instills pride	5	Subj.
18	13	Unit manufacturing cost	5	US\$
19	14	Time in spray chamber without water entry	5	s
20	15	Cycles in mud chamber without contamination	5	k-cycles
21	16, 17	Time to disassemble/assemble for maintenance	3	s
22	17, 18	Special tools required for maintenance	3	List
23	19	UV test duration to degrade rubber parts	5	hr
24	19	Monster cycles to failure	5	Cycles
25	20	Japan Industrial Standards test	5	Binary
26	20	Bending strength (frontal loading)	5	kN

EXHIBIT 6-4 List of metrics for the suspension. The relative importance of each metric and the units for the metric are also shown. "Subj." is an abbreviation indicating that a metric is subjective.

A few guidelines should be considered when constructing the list of metrics:

- **Metrics should be complete.** Ideally each customer need would correspond to a single metric, and the value of that metric would correlate perfectly with satisfaction of that need. In practice, several metrics may be necessary to completely reflect a single customer need.
- **Metrics should be dependent, not independent, variables.** This guideline is a variant of the *what-not-how principle* introduced in Chapter 5, Identifying Customer Needs. As do customer needs, specifications also indicate *what* the product must do, but not *how* the

Need	Metric																									
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1 Reduces vibration to the hands	•		•	•																						
2 Allows easy traversal of slow, difficult terrain		•																								
3 Enables high-speed descents on bumpy trails	•		•	•																						
4 Allows sensitivity adjustment					•																					
5 Preserves the steering characteristics of the bike						•	•																			
6 Remains rigid during hard cornering		•						•																		
7 Is lightweight									•																	
8 Provides stiff mounting points for the brakes										•																
9 Fits a wide variety of bikes, wheels, and tires											•	•	•	•												
10 Is easy to install															•											
11 Works with fenders																•										
12 Instills pride																	•									
13 Is affordable for an amateur enthusiast																		•								
14 Is not contaminated by water																			•							
15 Is not contaminated by grunge																				•						
16 Can be easily accessed for maintenance																					•					
17 Allows easy replacement of worn parts																					•	•				
18 Can be maintained with readily available tools																						•				
19 Lasts a long time																							•			
20 Is safe in a crash																								•	•	•

EXHIBIT 6-5 The needs-metrics matrix.

specifications will be achieved. Designers use many types of variables in product development; some are *dependent*, such as the mass of the fork, and some are *independent*, such as the material used for the fork. In other words, designers cannot control mass directly because it arises from other independent decisions the designers will make, such as dimensions and materials choices. Metrics specify the overall performance of a product and should therefore be the dependent variables (i.e., the performance measures or output variables) in the design problem. By using dependent variables for the specifications, designers are left with the freedom to achieve the specifications using the best approach possible.

Metrics should be practical. It does not serve the team to devise a metric for a bicycle suspension that can only be measured by a scientific laboratory at a cost of \$100,000. Ideally, metrics will be directly observable or analyzable properties of the product that can be easily evaluated by the team.

Some needs cannot easily be translated into quantifiable metrics. The need that the suspension instills pride may be quite critical to success in the fashion-conscious mountain bike market, but how can pride be quantified? In these cases, the team simply repeats the need statement as a specification and notes that the metric is subjective. (We indicate this by entering "Subj." in the units column.) However, the team still has to decide how to confirm that the specification is satisfied; for example, by a panel of customers.

The metrics should include the popular criteria for comparison in the marketplace. Many customers in various markets buy products based on independently published evaluations. Such evaluations are found, for example, in *Popular Science*, *Consumer Reports*, on various Internet sites, or, in our case, in *Bicycling* and *Mountain Bike* magazines. If the team knows that its product will be evaluated by the trade media and knows what the evaluation criteria will be, then it should include metrics corresponding to these criteria. *Mountain Bike* magazine uses a test machine called the Monster, which measures the vertical acceleration (in g's) of the handlebars as a bicycle equipped with the fork runs over a block 50 millimeters tall. For this reason, the team included "maximum value from the Monster" as a metric. If the team cannot find a relationship between the criteria used by the media and the customer needs it has identified, then it should ensure that a need has not been overlooked and/or should work with the media to revise the criteria. In a few cases, the team may conclude that high performance in the media evaluations is in itself a customer need and choose to include a metric used by the media that has little intrinsic technical merit.

In addition to denoting the needs related to each metric, Exhibit 6-4 contains the units of measurement and an importance rating for each metric. The units of measurement are most commonly conventional engineering units such as kilograms and seconds; however, some metrics will not lend themselves to numerical values. The need that the suspension "works with fenders" is best translated into a specification listing the models of fenders with which the fork is compatible. In this case, the value of the metric is actually a list of fenders rather than a number. For the metric involving the standard safety test, the value is pass/fail. (We indicate these two cases by entering "List" and "Binary" in the units column.)

The importance rating of a metric is derived from the importance ratings of the needs it reflects. For cases in which a metric maps directly to a single need, the importance rating of the need becomes the importance rating of the metric. For cases in which a metric is related to more than one need, the importance of the metric is determined by considering the importances of the needs to which it relates and the nature of these relationships. We believe that there are enough subtleties in this process that importance weightings can best be determined through discussion among the team members, rather than through a formal algorithm. When there are relatively few specifications and establishing the relative importance of these specifications is critically important, *conjoint analysis* may be useful. Conjoint analysis is described briefly later in this chapter and publications explaining the technique are referenced at the end of the chapter.

competitive Benchmarking Information

is to enjoy a total monopoly, the relationship of the new product to its paramount in determining commercial success. While the team product development process with some idea of how it wishes to place, the target specifications are the language the team uses to describe detailed positioning of its product relative to existing products, both its. Information on competing products must be gathered to support decisions.

competitive benchmarking chart is shown in Exhibit 6-6. The columns added to the competitive products and the rows are the metrics established that the competitive benchmarking chart can be constructed as a spreadsheet containing the list of metrics. (This information is one *house of Quality*, described by Hauser and Clausing.)

chart is conceptually very simple. For each competitive product, the chart is simply entered down a column. Gathering these data can be very taxing (at the least) purchasing, testing, disassembling, and estimating the most important competitive products; however, this investment in product development team can expect to succeed without having a word of warning: Sometimes the data contained in competitors' engineering literature are not accurate. Where possible, values of the key metrics by independent testing or observation.

competitive benchmarking chart can be constructed with rows corresponding to the competitive products and columns corresponding to the competitive products. The chart is used to compare customers' perceptions of the relative products satisfy their needs. Constructing this chart requires collecting data, which can also be very expensive and time consuming. Some engineering customers' perceptions of satisfaction of needs are contained in Hauser (1993). Both charts can be useful and any discrepancies are constructive. At a minimum, a chart showing the competitive values of the metrics should be created.

and Marginally Acceptable Target Values

synthesizes the available information to actually set the *target values* for the target value are useful: an *ideal value* and a *marginally acceptable value*. The first result the team could hope for. The marginally acceptable value is the value that would just barely make the product commercially viable. Both of these values define the subsequent stages of concept generation and concept selection, and the target specifications after the product concept has been selected.

to express the values of the metrics:

specifications establish targets for the lower bound on a metric, but not the upper bound. For example, the value of the brake mounting stiffness is specified in newtons/meter.

specifications establish targets for the upper bound on a metric, with the goal of being better. For example, the value for the mass of the suspension fork is 1.4 kilograms.

Metric	Imp.	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
ation ropout to bar at 10 Hz	3	dB	8	15	10	15	9	13
preload	3	N	550	760	500	710	480	680
um value ie Monster	5	g	3.6	3.2	3.7	3.3	3.7	3.4
um descent n test track	5	s	13	11.3	12.6	11.2	13.2	11
ng coefficient ment range	3	N-s/m	0	0	0	200	0	0
um travel wheel)	3	mm	28	48	43	46	33	38
ffset	3	mm	41.5	39	38	38	43.2	39
l stiffness at	3	kN/m	59	110	85	85	65	130
nass	4	kg	1.409	1.385	1.409	1.364	1.222	1.100
l stiffness at pivots	2	kN/m	295	550	425	425	325	650
set sizes	5	in.	1.000 1.125	1.000 1.125 1.250	1.000 1.125	1.000 1.125 1.250	1.000 1.125	NA
ube length	5	mm	150 180 210 230 255	140 165 190 215	150 170 190 210	150 170 190 210 230	150 190 210 220	NA
l sizes	5	List	26 in.	26 in.	26 in.	26 in. 700C	26 in.	26 in.

etitive benchmarking chart based on metrics.

between X and Y: These specifications establish both upper and lower bounds for the value of a metric. For example, the value for the spring preload is set to be between 480 and 800 newtons. Any more and the suspension is harsh; any less and the suspension is too bouncy.

Exactly X: These specifications establish a target of a particular value of a metric, with any deviation degrading performance. For example, the ideal value for the rake offset metric is set to 38 millimeters. This type of specification is to be avoided if possible because such specifications substantially constrain the design. Often, upon reconsideration, the team realizes that what initially appears as an "exactly X" specification can be expressed as a "between X and Y" specification.

A set of discrete values: Some metrics will have values corresponding to several discrete choices. For example, the headset diameters are 1.000, 1.125, or 1.250 inches. (Industry practice is to use English units for these and several other critical bicycle dimensions.)

Metric	Imp.	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
Maximum tire width	5	in.	1.5	1.75	1.5	1.75	1.5	1.5
Time to assemble to frame	1	s	35	35	45	45	35	85
Fender compatibility	1	List	Zefal	None	None	None	None	All
Installs pride	5	Subj.	1	4	3	5	3	5
Unit manufacturing cost	5	US\$	65	105	85	115	80	100
Time in spray chamber without re-entry	5	s	1300	2900	> 3600	> 3600	2300	> 3600
Revolutions in mud chamber without re-amination	5	k-cycles	15	19	15	25	18	35
Time to disassemble/ re-able for re-ance	3	s	160	245	215	245	200	425
Tools needed for re-ance	3	List	Hex	Hex	Hex	Hex	Long hex	Hex, pin wrench
Duration to rubber parts	5	hr	400+	250	400+	400+	400+	250
Cycles to	5	Cycles	500k+	500k+	500k+	480k	500k+	330k
Industrial test	5	Binary	Pass	Pass	Pass	Pass	Pass	Pass
Strength (kg)	5	kN	5.5	8.9	7.5	7.5	6.2	10.2

able range of values for one metric may depend on another. In other words, we may set a target as, for example, "the fork tip lateral stiffness is no more than 20 percent stiffness at the brake pivots." In applications where the team feels this level of warranted, such targets can easily be included, although we recommend that this complexity not be introduced until the final phase of the specifications process.

five different types of expressions for values of the metrics, the team sets specifications. The team simply proceeds down the list of metrics and assigns naturally acceptable and ideal target values for each metric. These decisions by the metric-based competitive benchmarking chart shown in Exhibit 6-6. values, the team has many considerations, including the capability of competing at the time, competitors' future product capabilities (if these are predictable), mission statement and target market segment. Exhibit 6-8 shows the targets for suspension fork.